

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-ac3d737943ecaabfa.jsonl`  
- SHA-256: `9b2fdda90a1f84127e779090135e1951cfbc5318375d2c7577248fe6287bcf92`  
- Source modified: `2026-06-06T17:31:54+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:30:55.777Z)

Source Extractor

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Fetch and extract key claims from this source:

****URL:**** <https://ar5iv.labs.arxiv.org/html/2204.01899>

****Title:**** MonoTrack: Shuttle Trajectory Reconstruction from Monocular Badminton Video (HitNet)

****Found via:**** practitioner/local-levers search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:30:58.925Z)

I'll fetch the source and extract the relevant claims.

3. user (2026-06-06T17:31:15.550Z)

MonoTrack: Key Claims Extraction

(1) HitNet Module for Hit Detection

****Architecture & Inputs:****

"recurrent model that leverages the previously computed court layout, poses, and shuttlecock positions and their temporal tracks to predict hits"

The model uses "a feature vector comprising of the pixel coordinates of the court corners, the two players' poses, and the location of the shuttle" fed through "two GRU layers that takes 12 frames at a time."

****Accuracy:****

"The network is compact in size (around 16K parameters) and can reach 86.3% accuracy." With constrained optimization applied: "HitNet+optimization" achieves "94.3% recall, 76.4% accuracy, 97.2% precision."

(2) Shuttle Trajectory Reconstruction

****Method:****

"Physics-based trajectory estimation" using "particle under drag" model solved via "constrained nonlinear optimization problem."

****Accuracy:****

On synthetic data: "adding the priors improves the error from 14.9 cm to 8.0 cm." On real data with ground truth: "8.1 px, 8.9 px, 9.8 px" reprojection error across three matches.

(3) Rally/Point Segmentation via Hit Detection

"shot happens when a player hits the shuttle with her racket, and ends right before the opposing player hits the shuttle...therefore, in order to segment the video into successive shots, it is equivalent to identify the hit events."

(4) Datasets and Licenses

"dataset is based on the public TrackNetV2 dataset...contains 77k annotated frames from 26

unique singles matches."

****License:**** Not explicitly stated in the document.

(5) Code/Model License

Not mentioned in the paper.

(6) Compute Requirements

"2.9M parameters" for shuttle detection model vs. "11.3M parameters" in TrackNetV2. "performance inference over several thousand frames a second" for HitNet.

(7) Publication Details

arXiv preprint 2204.01899; appears to be 2022 based on identifier.

4. assistant (2026-06-06T17:31:18.116Z)

Let me get more precise details on the dataset/license, the stroke-type classification, and the publication metadata to ground the claims.

5. user (2026-06-06T17:31:28.393Z)

MonoTrack: Shuttle Trajectory Reconstruction - Key Details

(a) Hit Detection Accuracy (HitNet)

The paper reports hit detection performance in Table 1:

****HitNet + optimization:**** "94.3% recall, 76.4% accuracy, 97.2% precision, 0.946 f1"

****HitNet (All features):**** "78.1% recall, 76.4% accuracy, 97.2% precision, 0.866 f1"

The baseline comparison shows: "Derivative-based: 65.7% recall, 53.8% accuracy, 74.7% precision, 0.699 f1"

No direct TrackNetV2 hit-detection comparison is provided in the paper.

(b) TrackNetV2 Dataset Details

"This dataset contains 77k annotated frames from 26 unique singles matches from international play, filmed from an overhead, static, 'broadcast-view' camera."

The authors note the dataset is "public" and they "use the 12k frames from their three 'test matches' for testing."

(c) Shuttle Detection Accuracy vs TrackNetV2

"Our model has significantly smaller footprint than the original TrackNetV2 (2.9M parameters in our model versus 11.3M parameters), making it faster to train and perform inference, and higher accuracy (88.6% accuracy from 84.0% in the original)."

No precision/recall/F1 metrics are reported for shuttle detection.

(d) Publication Details

The document header shows: "[2204.01899]" (arXiv identifier). No conference venue, submission date, or publication date is explicitly stated in the provided content.

(e) Title and Authors

****Title:**** "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video"

****Authors:**** Paul Liu (Stanford University) and Jui-Hsien Wang (Adobe Research, Seattle)

6. assistant (2026-06-06T17:31:31.889Z)

I have enough to extract the claims. The publish date for arXiv 2204.01899 corresponds to April 2022 (the YYYY identifier convention: 22=2022, 04=April). The paper appeared at CVPR Workshops (CVSports) 2022, but since that isn't confirmed verbatim in the fetched content, I'll report the date conservatively.

7. user (2026-06-06T17:31:44.939Z)

Structured output provided successfully

8. assistant (2026-06-06T17:31:54.160Z)

Extracted 5 falsifiable claims from the MonoTrack paper (arXiv 2204.01899, Liu & Wang, Stanford/Adobe Research ? a primary research source).

Key takeaways most relevant to the research question:

- HitNet directly validates the pose + trajectory + court-geometry hit-detection approach that our pipeline can run fully locally, and explicitly frames hit detection as the segmentation primitive (hits = shot/rally boundaries).
- Its learned model (0.946 F1) substantially beats velocity/derivative-threshold heuristics (0.699 F1) ? strong evidence that our current threshold-based WASB windowing is leaving accuracy on the table, and that a small learned hit detector is a competitive, local-only alternative/complement to the Gemini-teacher distillation path.
- It is tiny (~16K params, thousands of fps), so it fits the RTX 2070 Super budget easily.
- Caveat flagged: the paper does not state a code/weights license, so MonoTrack's commercial usability is unconfirmed (the underlying TrackNetV2 data is public). This matches prior project notes about avoiding MonoTrack weights on licensing grounds.

One uncertainty: the exact publish date and venue are not verbatim in the ar5iv content; the arXiv ID 2204 indicates April 2022.